

G. NARAYANAMMA INSTITUTE OF TECHNOLOGY & SCIENCE
Autonomous **(For Women)**

Shaikpet, Hyderabad - 500104

DEPARTMENT OF CSE
III B. Tech I Sem (DS Minor)
R PROGRAMMING LABORATORY

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	give names to the elements in the list and display the list also access the list elements iii. To add a new element at the end of the list and delete the element from the middle display the same iv. To create two lists, merge two lists. Convert the lists into vectors and perform addition on the two vectors. Display the resultant vector.	
9	Write R command for i. logical operators—AND (&), OR () and NOT (!). ii. Conditional Statements iii. Create four vectors namely patientid, age, diabetes, and status. Put these four vectors into a data frame patientdata and print the values using a for loop & While loop iv. Create a user-defined function to compute the square of an integer in R v. Create a user-defined function to compute the square of an integer in R vi. Recursion function for a) factorial of a number b) find nth Fibonacci number	27
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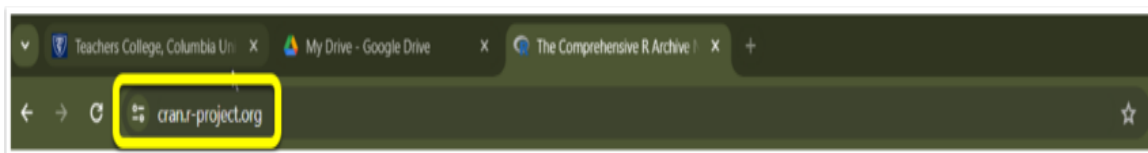
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1. R Environment setup: Installation of R and RStudio in Windows

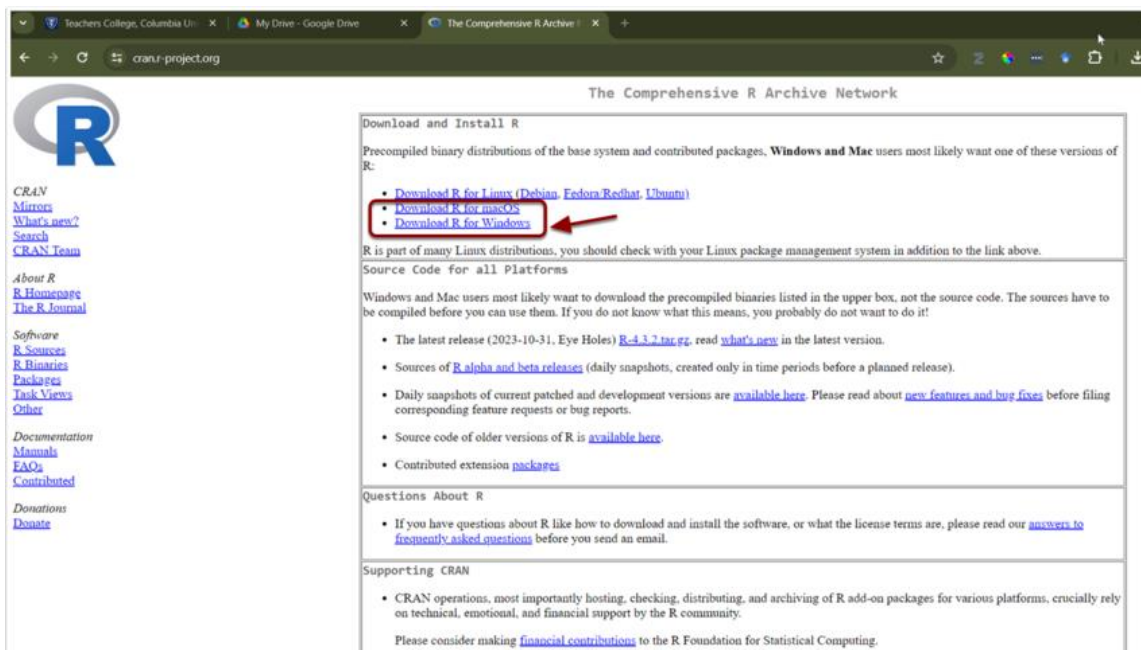
Steps:

1. To install R, go to cran.r-project.org

cran.r-project.org



2. Choose for Download R for Windows

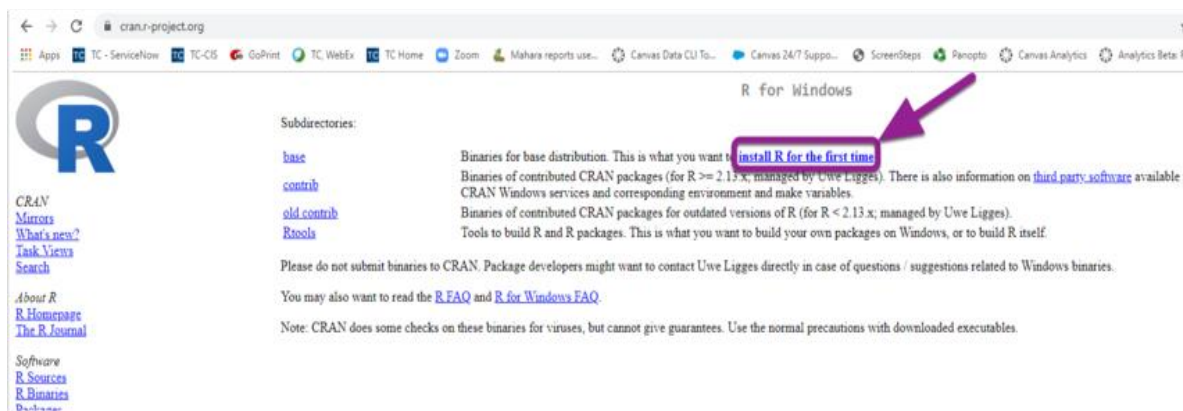


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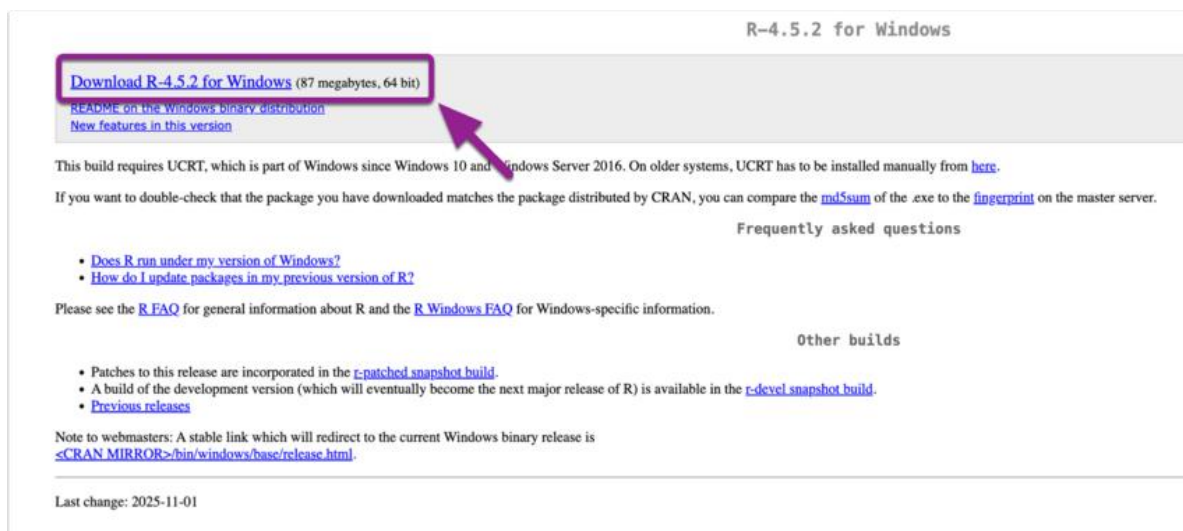
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3. Install R Click on install R for the first time.

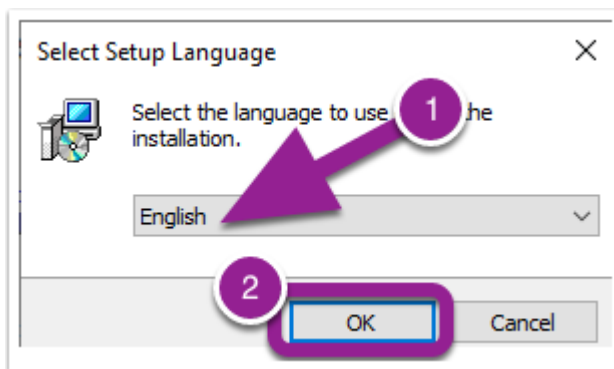


4. Click Download R for Windows. Open the downloaded file



5. Select the language you would like to use during the installation.

Then click OK

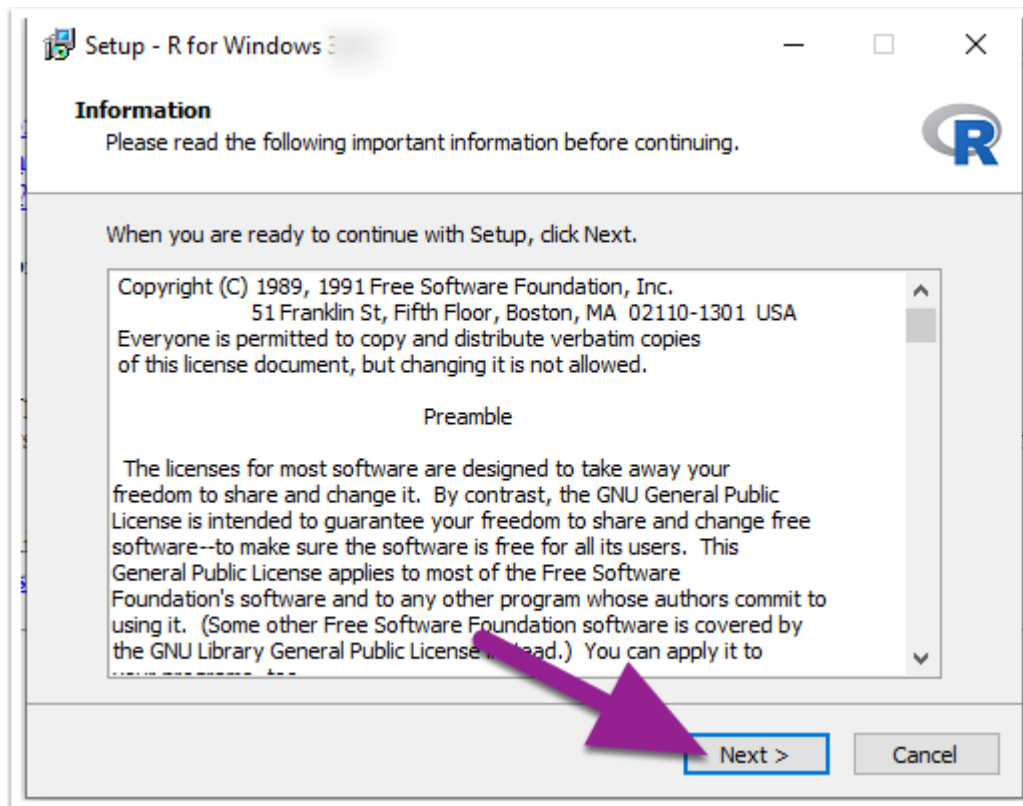


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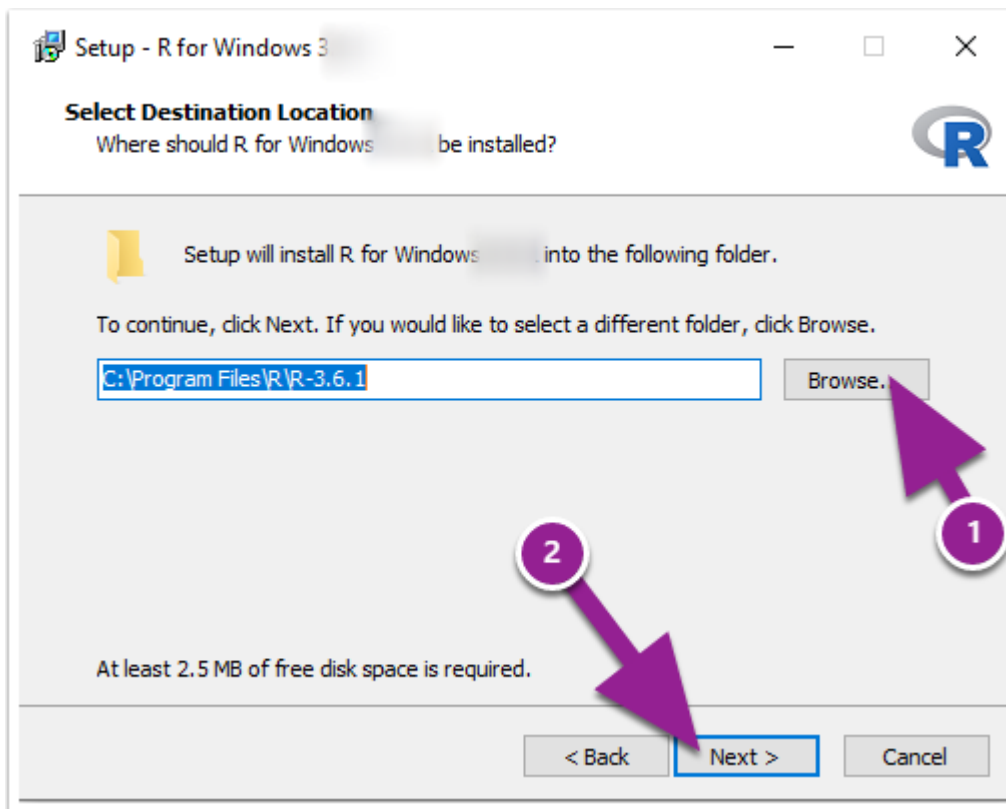
6. Click Next.



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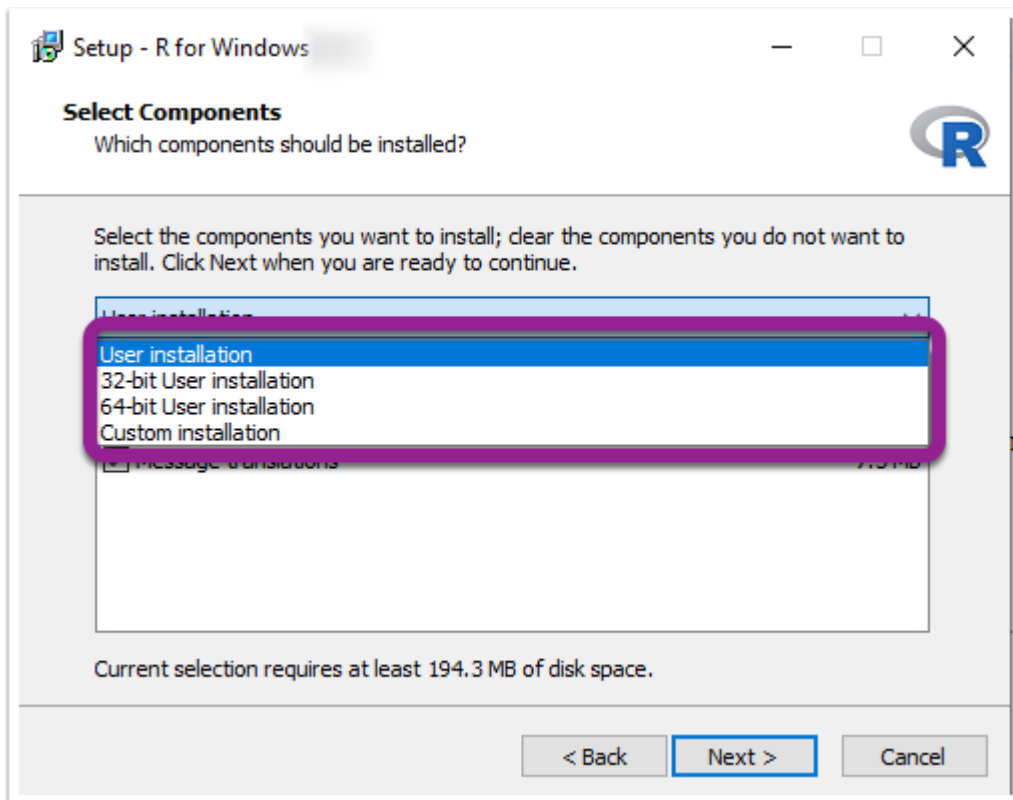
7. Select where you would like R to be installed.

It will default to your Program Files on your C Drive. Click Next.



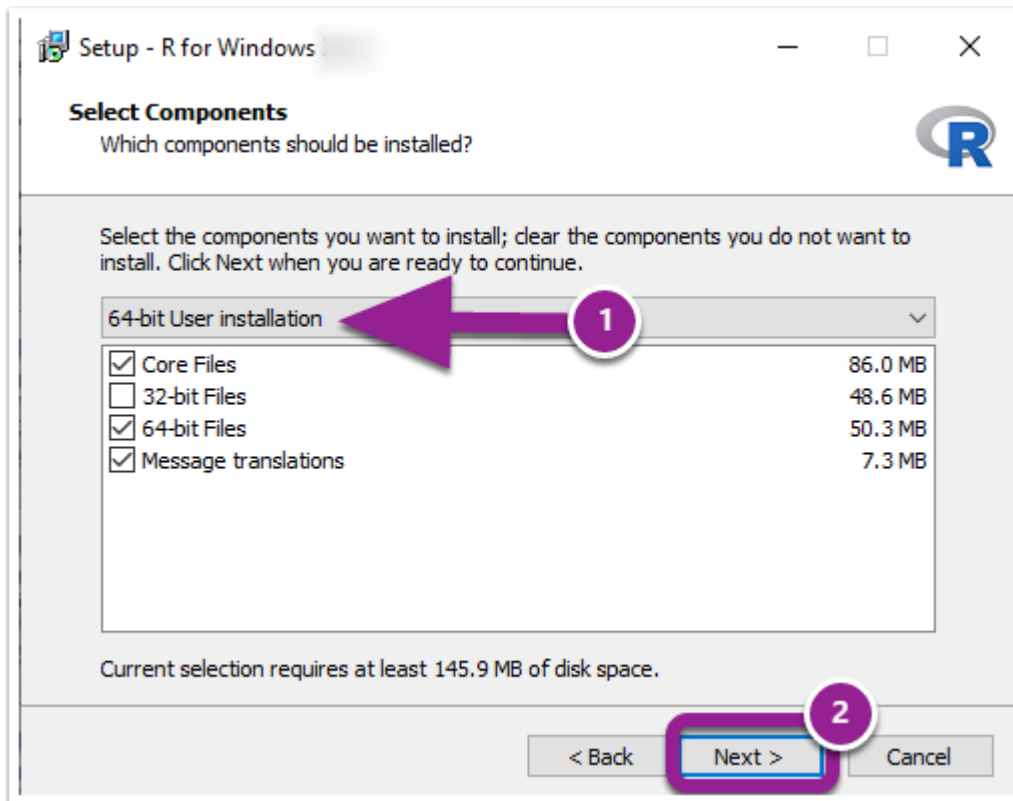
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8. You can then choose which installation you would like.

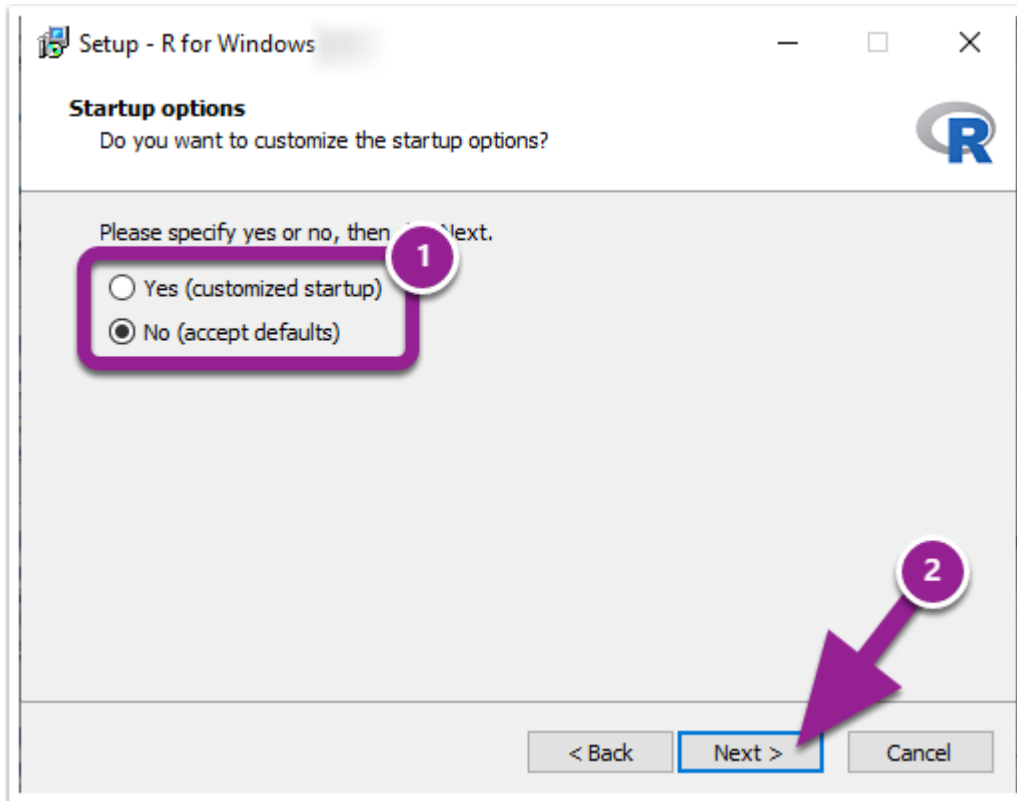


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9. (Optional) If your computer is a 64-bit, you can choose the 64-bit User Installation. Then click Next.



10. Then specify if you want to customized your startup or just use the defaults. Then click Next.



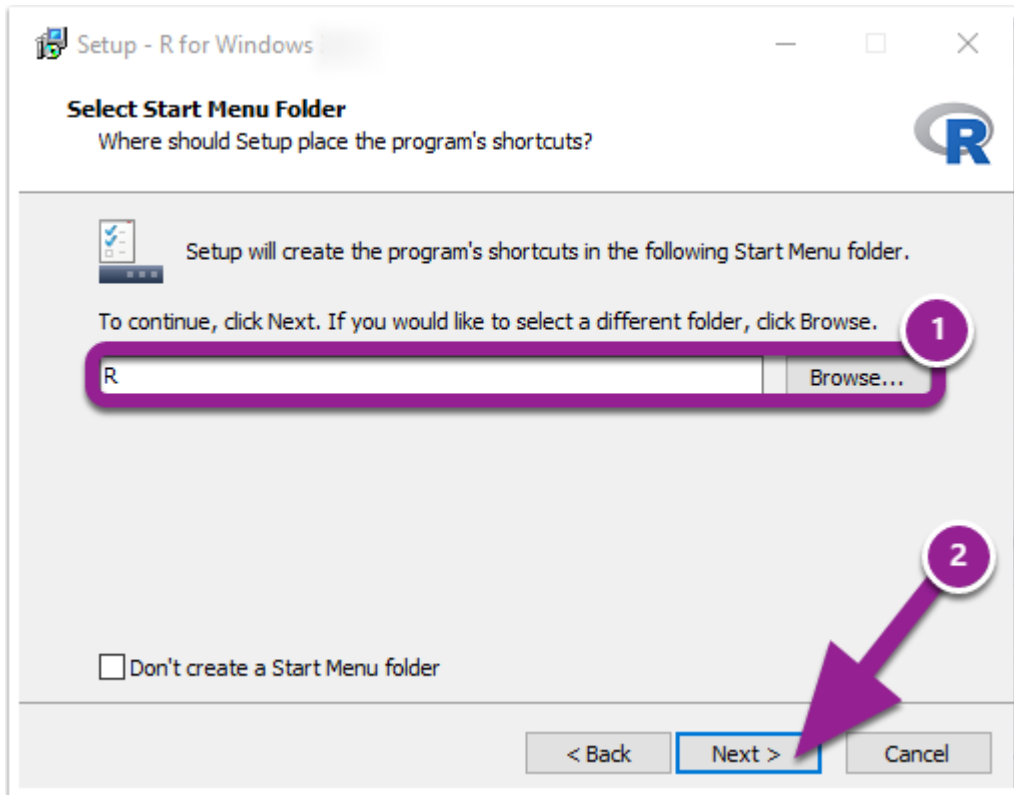
11. Then you can choose the folder that you want R to be saved within or the default if the R folder that was created. Once you have finished, click Next.

You can also choose if you do not want a Start Menu folder at the bottom.

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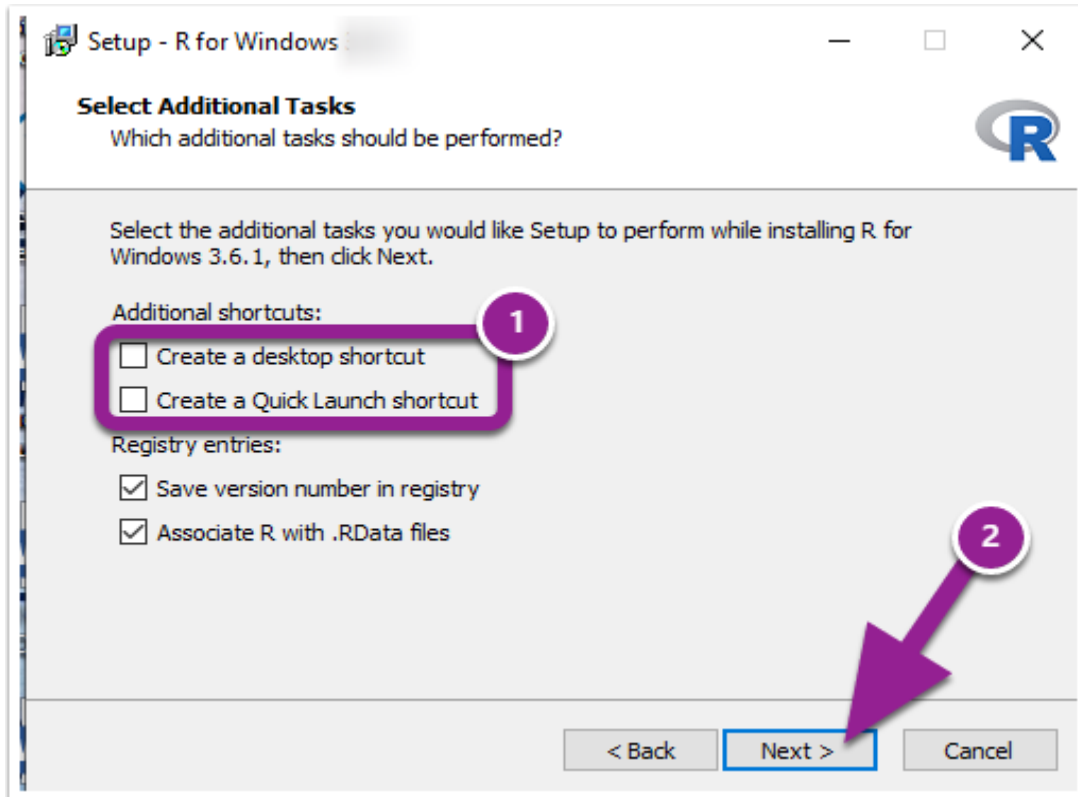
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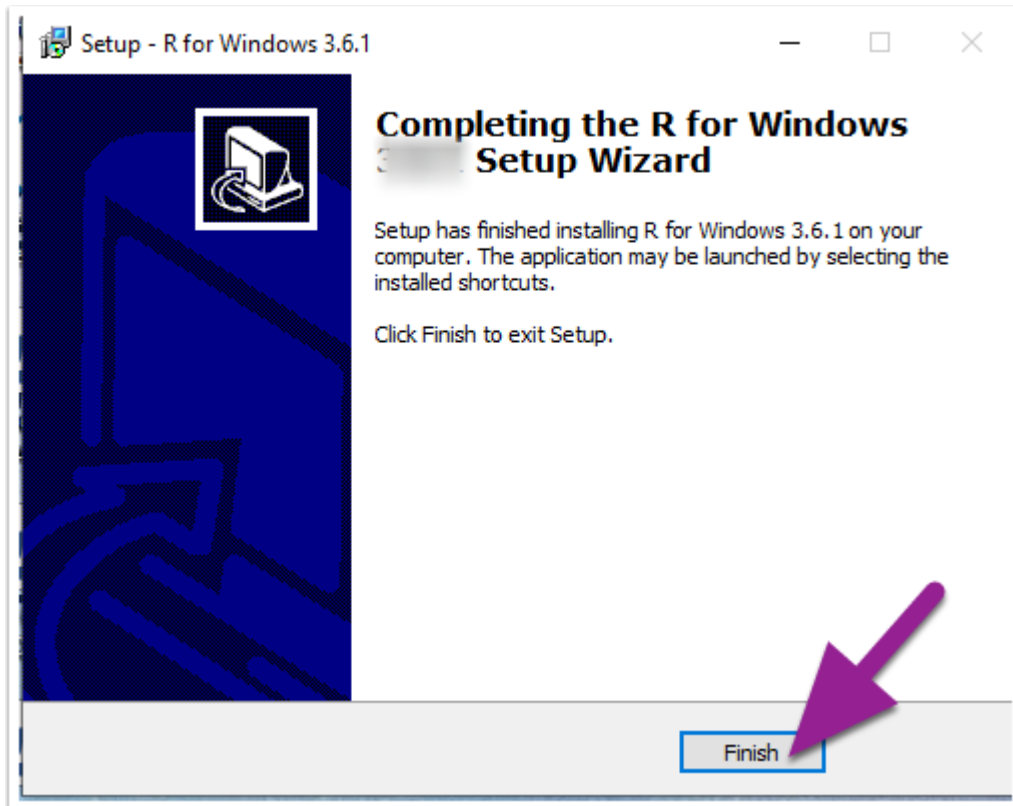


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12. You can then select additional shortcuts if you would like. Click Next.

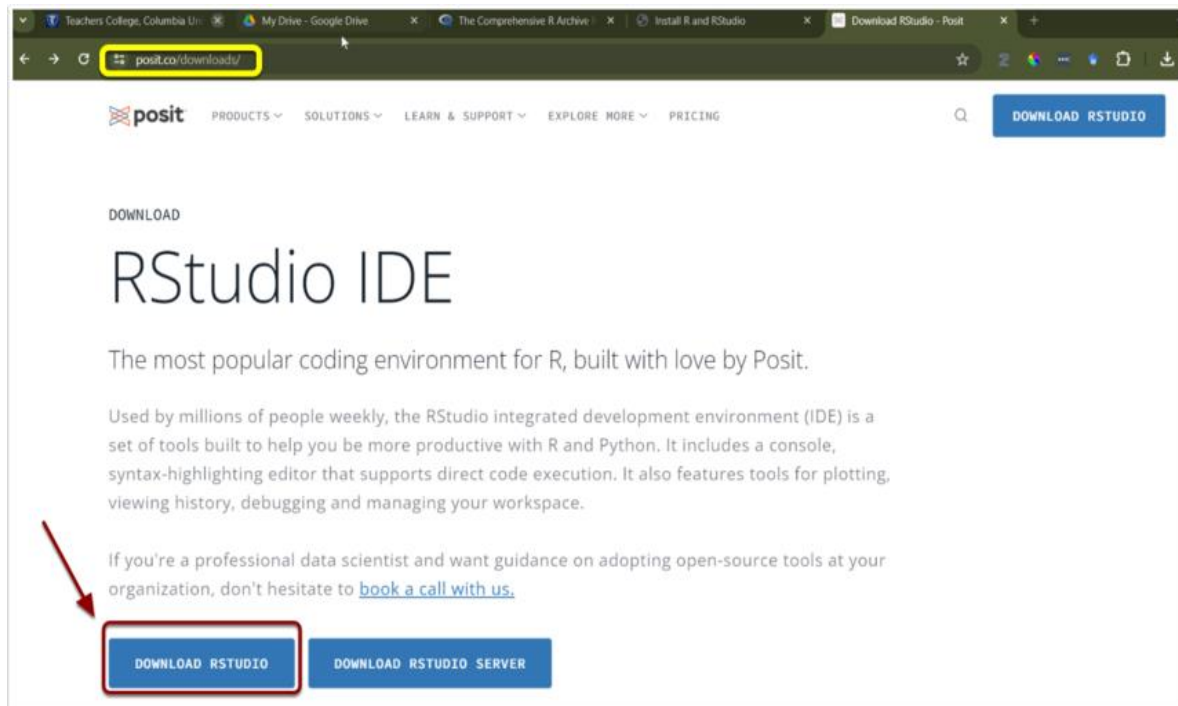


13. Click Finish.



14. Next, download RStudio

Go to <https://posit.co/downloads/>



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15. Click Download RStudio

Teachers College, Columbia Uni x My Drive - Google Drive x The Comprehensive R Archive x Install R and RStudio x RStudio Desktop - Posit x

posit.co/download/rstudio-desktop/

posit PRODUCTS SOLUTIONS LEARN & SUPPORT EXPLORE MORE PRICING

Don't want to download or install anything? Get started with RStudio on [Posit Cloud for free](#). If you're a professional data scientist looking to download RStudio and also need common enterprise features, don't hesitate to [book a call with us](#).

1: Install R

RStudio requires R 3.3.0+. Choose a version of R that matches your computer's operating system.

[DOWNLOAD AND INSTALL R](#)

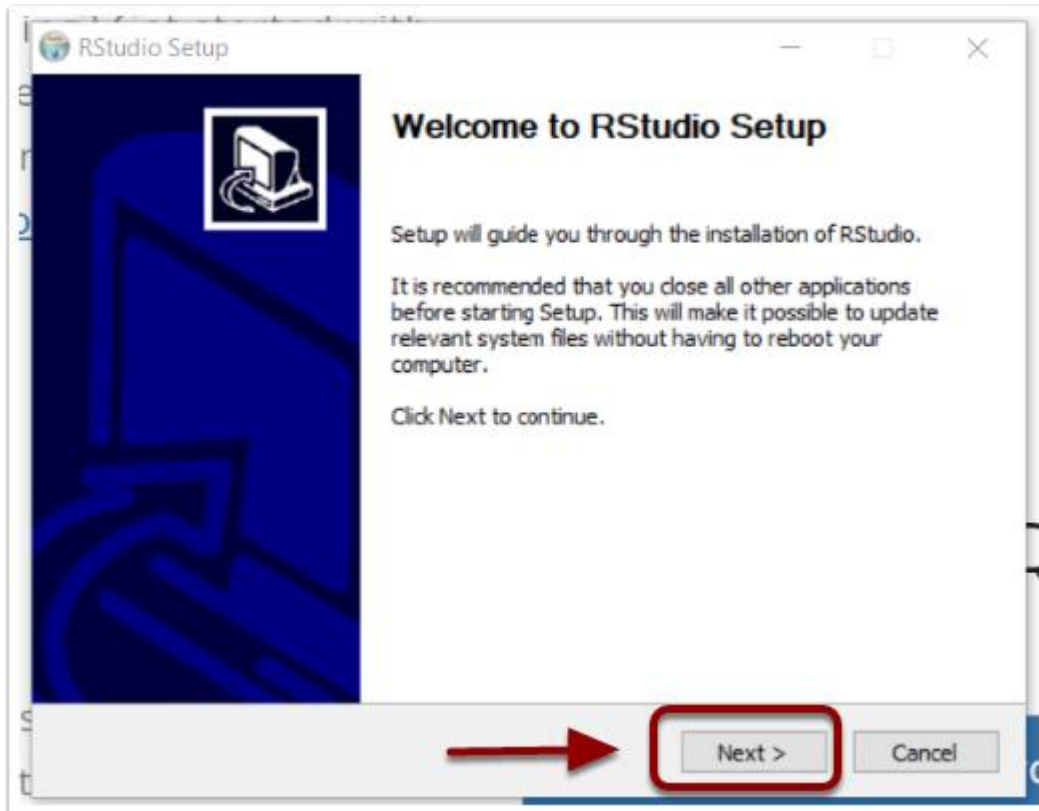
2: Install RStudio

[DOWNLOAD RSTUDIO DESKTOP FOR WINDOWS](#)

Size: 214.34 MB | [SHA-256: FE62B784](#) | Version: 2023.09.1+494 | Released: 2023-10-17

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16. Once the packet has downloaded, the Welcome to RStudio Setup Wizard will open. Click Next and go through the installation steps.

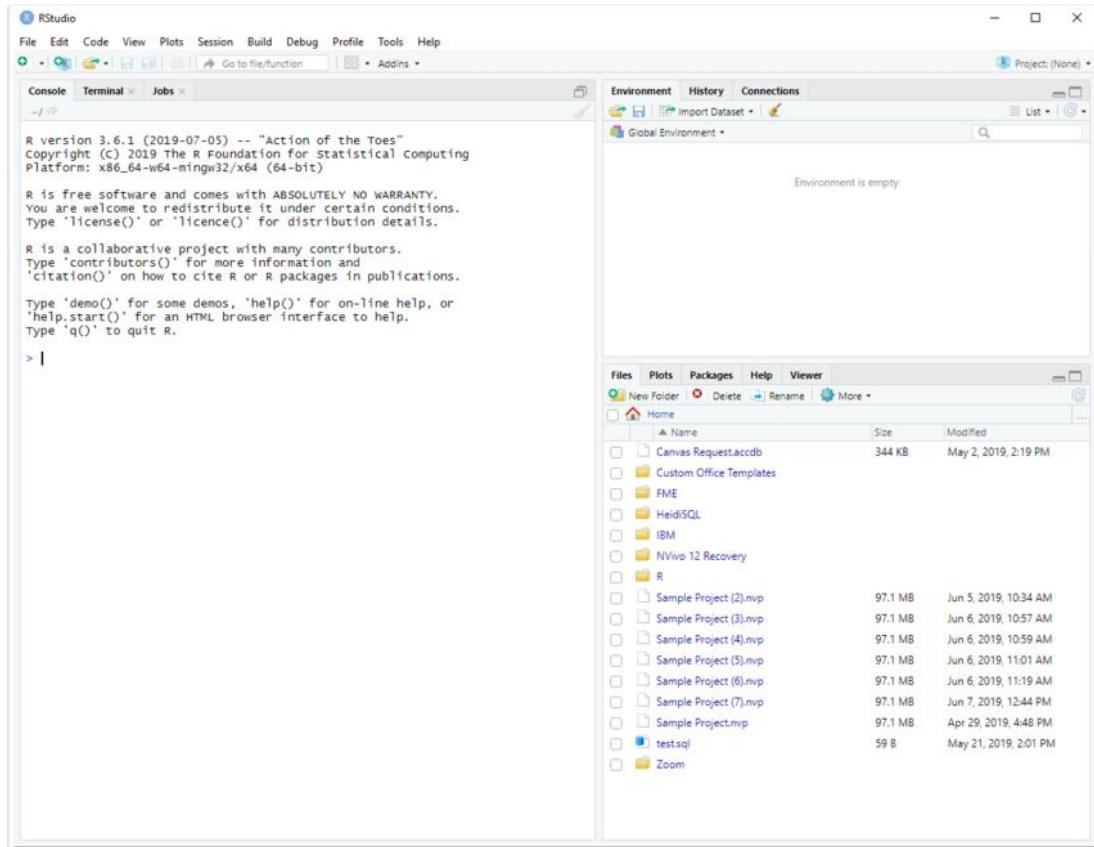


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17. After the Setup Wizard finishing the installation, RStudio will open.



2. Write R commands for

i. Variable declaration and retrieving the value of the stored variables,

ii. Write an R script with comments,

iii. Type of a variable using class () Function.

2 i) PROGRAM:

```
# Variable declaration
```

```
x <- 10
```

```
y <- 20
```

```
# Retrieving values
```

```
x
```

```
y
```

Output:

```
[1] 10
```

```
[1] 20
```

2 ii) PROGRAM:

```
# This is a comment
```

```
a <- 5 # assigning value
```

```
b <- 3
```

```
# Adding values
```

```
sum <- a + b
```

```
sum
```

Output:

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```
[1] 8
```

2 iii) PROGRAM

```
x <- 10  
y <- "Hello"  
z <- TRUE
```

```
class(x)  
class(y)  
class(z)
```

Output:

```
[1] "numeric"  
[1] "character"  
[1] "logical"
```

3. Write R command to

- i. illustrate summation, subtraction, multiplication, and division operations on vectors using vectors.**
- ii. Enumerate multiplication and division operations between matrices and vectors in R console**

3 i) PROGRAM

```
v1 <- c(1,2,3)  
v2 <- c(4,5,6)
```

```
v1 + v2  
v1 - v2
```

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`v1 * v2`

`v1 / v2`

Output:

`[1] 5 7 9`

`[1] -3 -3 -3`

`[1] 4 10 18`

`[1] 0.25 0.40 0.50`

3 ii) PROGRAM

`m <- matrix(c(1,2,3,4), nrow=2)`

`v <- c(2,2)`

`m * v`

`m / v`

Output:

`[,1] [,2]`

`[1,] 2 6`

`[2,] 4 8`

`[,1] [,2]`

`[1,] 0.5 1.5`

`[2,] 1.0 2.0`

6. Write R command to

i. Illustrate the usage of Vector sub setting& Matrix sub setting

ii. Write a program to create an array of 3×3 matrixes with 3 rows and 3 columns.

iii. Write a program to create a class, object, and function

6i) PROGRAM:

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```
v <- c(10,20,30,40)
```

```
v[2]
```

```
m <- matrix(1:9, nrow=3)
```

```
m[2,3]
```

Output:

```
[1] 20
```

```
[1] 8
```

6ii) PROGRAM

```
arr <- array(1:27, dim=c(3,3,3))
```

```
arr
```

Output:

3 matrices printed (1–27 arranged in 3 layers)

6iii) PROGRAM

```
#We define a "constructor" function that creates a list and assigns it a class
```

```
Person <- function(name, age, occupation) {
```

```
# Create a list containing the object's data
```

```
obj <- list( name = name, age = age, occupation = occupation )
```

```
# Assign a class name to this object
```

```
class(obj) <- "Person" return(obj) }
```

```
# Create an object
```

```
# Instantiate the 'Person' class
```

```
my_object <- Person(name = "Alice", age = 30, occupation = "Data Scientist")
```

```
# Output the results
```

```
print(my_object) print(my_object$name)
```

```
# Check the class of the object
```

```
print(paste("The object class is:", class(my_object)))
```

Output:

```
$name  
[1] "Alice"
```

```
$age  
[1] 30
```

```
$occupation  
[1] "Data Scientist"
```

```
attr(,"class")  
[1] "Person"  
[1] "Alice"  
[1] "The object class is: Person"
```

7. Write a command in R console

i. to create a `tshirt_factor`, which is ordered with levels 'S', 'M', and 'L'. Is it possible to identify from the examples discussed earlier, if blood type 'O' is greater or less than blood type 'A'?

ii. Write the command in R console to create a new data frame containing the 'age' parameter from the existing data frame. Check if the result is a data frame or not. Also R commands for data frame functions `cbind()`, `rbind()`, `sort()`

7i) PROGRAM

```
tshirt_factor <- factor(c("S","M","L"), ordered=TRUE, levels=c("S","M","L"))  
tshirt_factor
```

Output:

```
[1] S M L  
Levels: S < M < L
```

7 ii) PROGRAM:

```
df <- data.frame(name=c("A","B"), age=c(20,25))  
  
new_df <- df["age"]  
class(new_df)  
  
cbind(df, gender=c("M","F"))  
rbind(df, data.frame(name="C", age=30))  
sort(df$age)
```

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Output:

```
[1] "The object class is: Person"
[2] "data.frame"
```

	name	age	gender
1	A	20	M
2	B	25	F

	name	age
1	A	20
2	B	25
3	C	30

```
[1] 20 25
```

8. Write R command for

- Create a list containing strings, numbers, vectors and logical values
- To create a list containing a vector, a matrix, and a list. Also give names to the elements in the list and display the list also access the list elements
- To add a new element at the end of the list and delete the element from the middle display the same
- To create two lists, merge two lists. Convert the lists into vectors and perform addition on the two vectors. Display the resultant vector.

8i) PROGRAM:

```
lst <- list("Hello", 10, c(1,2), TRUE)
lst
```

OUTPUT:

```
[[1]]
[1] "Hello"
```



```
[[2]]  
[1] 10
```

```
[[3]]  
[1] 1 2
```

```
[[4]]  
[1] TRUE
```

8ii) PROGRAM

```
lst <- list(vec=c(1,2), mat=matrix(1:4,2), sublist=list(a=1))
```

```
lst
```

```
lst$vec
```

OUTPUT

```
$vec
```

```
[1] 1 2
```

```
$mat
```

```
      [,1] [,2]
```

```
[1,]     1     3
```

```
[2,]     2     4
```

```
$sublist
```

```
$sublist$a
```

```
[1] 1
```

```
> lst$vec
```

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```
[1] 1 2
```

8iii) PROGRAM

```
lst[[4]] <- "New"
```

```
lst[[2]] <- NULL
```

```
lst
```

OUTPUT

```
$vec
```

```
[1] 1 2
```

```
$sublist
```

```
$sublist$a
```

```
[1] 1
```

```
[[3]]
```

```
[1] "New"
```

8iv) PROGRAM

```
l1 <- list(1,2,3)
```

```
l2 <- list(4,5,6)
```

```
merged <- c(l1,l2)
```

```
v1 <- unlist(l1)
```

```
v2 <- unlist(l2)
```

```
v1 + v2
```

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OUTPUT

```
$vec
```

```
[1] 1 2
```

```
$sublist
```

```
$sublist$a
```

```
[1] 1
```

```
[[3]]
```

```
[1] "New"
```

9. Write R command for i. logical operators—AND (&), OR (|) and NOT (!). ii. Conditional Statements iii. Create four vectors namely patientid, age, diabetes, and status. Put these four vectors into a data frame patientdata and print the values using a for loop & While loop iv. Create a user-defined function to compute the square of an integer in R v. Create a user-defined function to compute the square of an integer in R vi. Recursion function for a) factorial of a number b) find nth Fibonacci number

9i) PROGRAM:

```
a <- TRUE; b <- FALSE
```

```
a & b; a | b; !a
```

OUTPUT:

```
[1] FALSE
```

```
[1] TRUE
```

```
[1] FALSE
```

9ii) PROGRAM:

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```
a <- TRUE; b <- FALSE
a & b; a | b; !a
x <- 10
if(x>5){
print("Greater")
} else {print("Smaller")}
```

OUTPUT:

```
[1] "Greater"
```

9iii) PROGRAM:

```
patientid <- c(1,2,3)
age <- c(25,30,35)
diabetes <- c("Yes","No","Yes")
status <- c("Good","Bad","Good")
patientdata <- data.frame(patientid, age, diabetes, status)

for(i in 1:nrow(patientdata)){
  print(patientdata[i,])
}

i <- 1
while(i <= nrow(patientdata)){
  print(patientdata[i,])
  i <- i + 1
}
```

OUTPUT:

```
patientid age diabetes status
1         1    25      Yes   Good
patientid age diabetes status
2         2    30       No    Bad
patientid age diabetes status
3         3    35      Yes   Good
```

9iv) PROGRAM:

```
square <- function(x){x^2}  
square(5)
```

OUTPUT:

```
[1] 25
```

9v) PROGRAM:

```
square <- function(x){x^2}  
square(5)
```

OUTPUT:

```
[1] 25
```

9vi) PROGRAM:

```
a) fact <- function(n){  
    if(n==0) return(1)    else return(n*fact(n-1))}  
    fact(5)
```

```
b) fib <- function(n){  
    if(n<=1) return(n)  
    else return(fib(n-1)+fib(n-2))  
}  
fib(6)
```

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OUTPUT:

a) [1] 120
b) [1] 8

10. Write R code for

i) Illustrate Quick Sort

ii) Illustrate Binary Search Tree

10i) PROGRAM:

```
quick_sort <- function(x){  
  if(length(x)<=1) return(x)  
  pivot <- x[1]  
  c(quick_sort(x[x<pivot]), pivot, quick_sort(x[x>pivot]))  
}  
quick_sort(c(5,3,8,2))
```

OUTPUT:

[1] 2 3 5 8

10ii) PROGRAM:

```
binary_search <- function(arr,x){  
  low<-1; high<-length(arr)  
  while(low<=high){  
    mid<-floor((low+high)/2)  
    if(arr[mid]==x) return(mid)  
    else if(arr[mid]<x) low<-mid+1  
    else high<-mid-1  
  }  
  return(-1)  
}
```

```
}  
binary_search(c(1,2,3,4,5),4)
```

OUTPUT:

```
[1] 4
```

11. Write R command to

i. illustrate Mathematical functions & I/O functions

ii. Illustrate Naming of functions and `sapply()`, `lapply()`, `tapply()` & `mapply()`

11i) PROGRAM:

```
# Basic math operations  
x <- -16.7  
abs_val <- abs(x) # 16.7  
sq_rt <- sqrt(16) # 4  
rounded <- round(3.1415, 2) # 3.14  
# Vector operations  
vec <- c(1, 2, 3, 4)  
total_sum <- sum(vec) # 10  
total_prod <- prod(vec) # 24  
cat(x, abs_val, sq_rt, rounded)
```

OUTPUT:

```
-16.7 16.7 4 3.14
```

11ii) PROGRAM:

```
x <- list(1:3,4:6)  
lapply(x,sum)  
sapply(x,sum)  
tapply(c(1,2,3,4), c("A","A","B","B"), sum)  
mapply(sum,1:3,4:6)
```

OUTPUT:

```
[[1]]  
[1] 6
```

```
[[2]]  
[1] 15
```

```
[1] 6 15
```

```
A B  
3 7
```

```
[1] 5 7 9
```

12. Write R command for

- i. Pie chart & 3D Pie Chart, Bar Chart to demonstrate the percentage conveyance of various ways for traveling to office such as walking, car, bus, cycle, and train
- ii. Using box plots, Histogram, Line Graph, Multiple line graphs and scatter plot to demonstrate the relation between the cars speed and the distance taken to stop,

12i) PROGRAM:

```
values <- c(10,20,30,15,25)
```

```
labels <- c("Walking","Car","Bus","Cycle","Train")
```

```
cols <- c("red","blue","yellow","green","white")
```

```
pie(values, labels=labels, col=cols)
```

```
legend("topright", legend=labels, fill=cols)
```

```
barplot(values, col=cols)
```

```
plot(cars$speed, cars$dist)
```

```
hist(cars$speed)
```

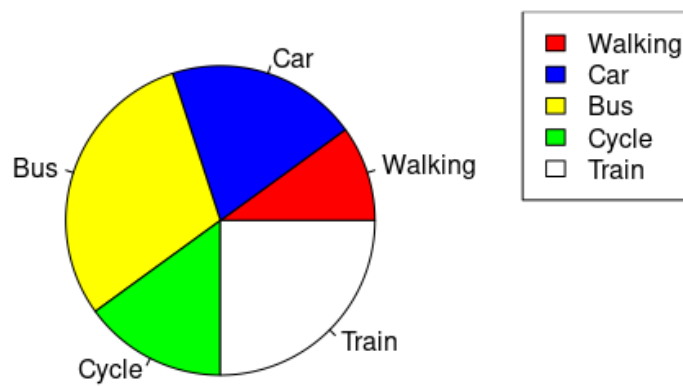


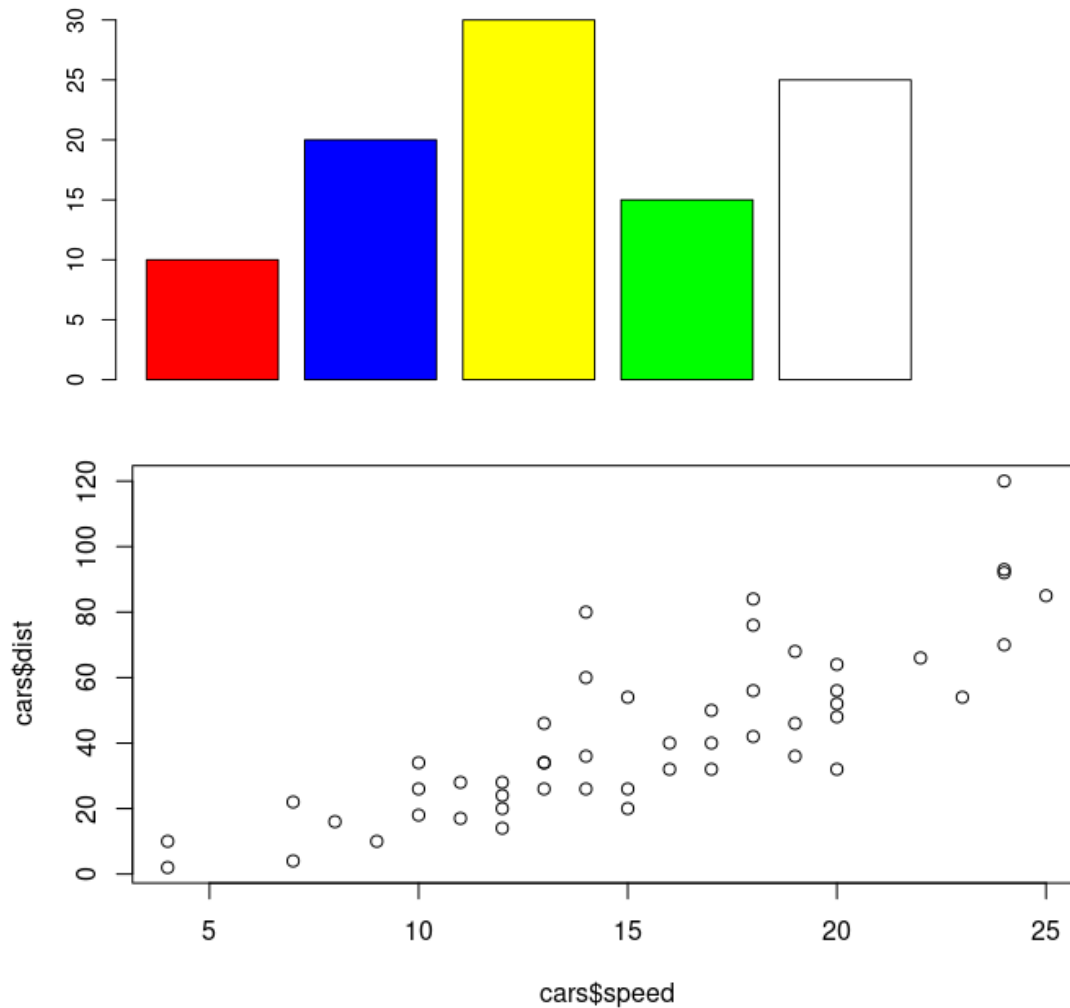
```
boxplot(cars$dist)
```

```
plot(cars$speed, cars$dist, type="l")
```

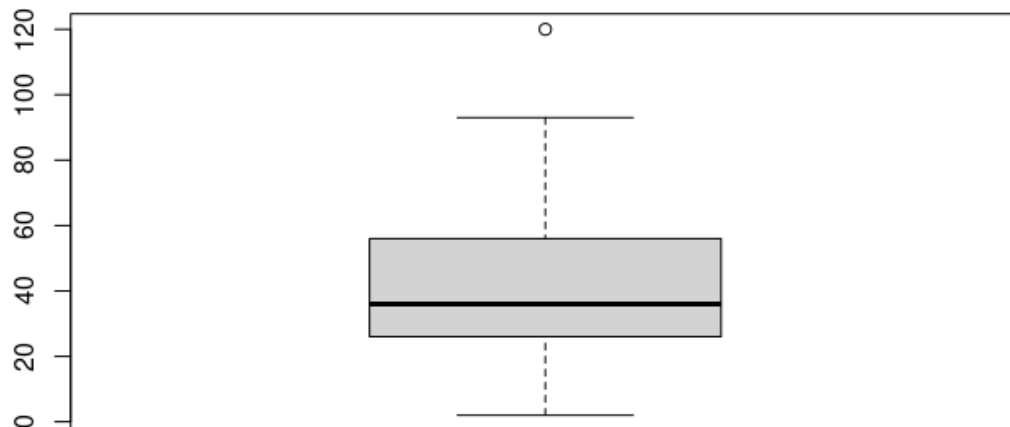
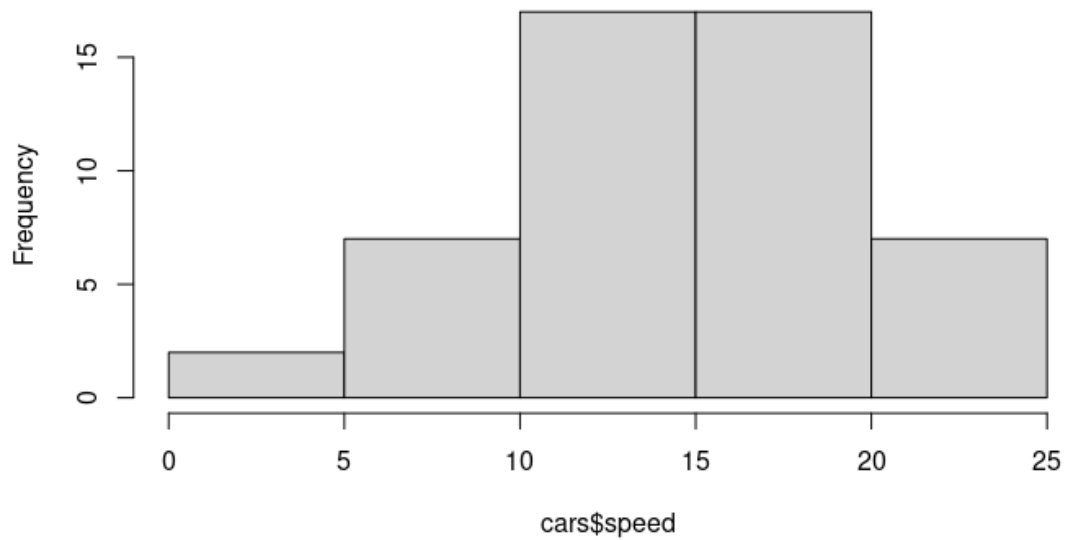
```
plot(cars$speed, cars$dist, col="blue")
```

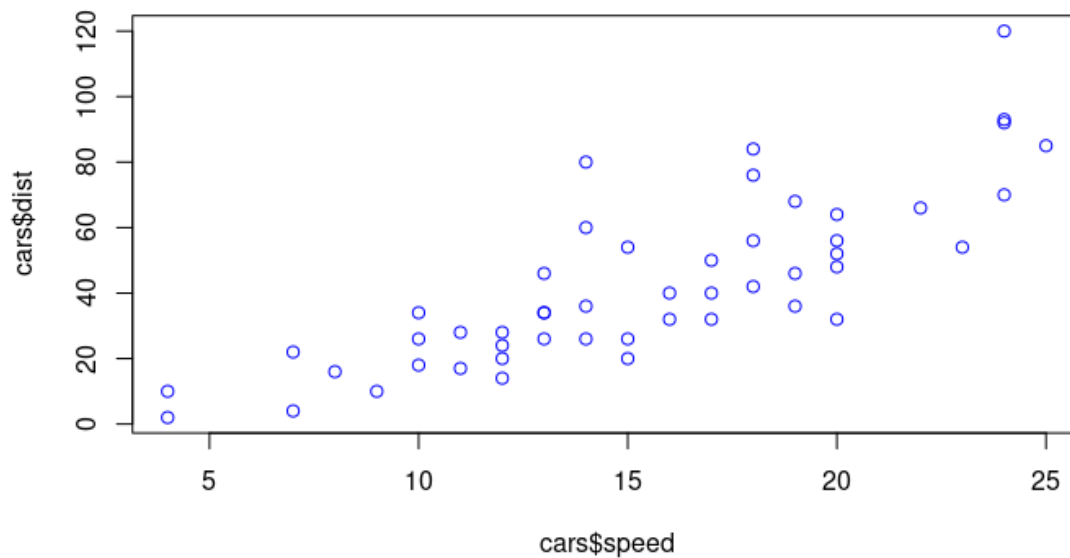
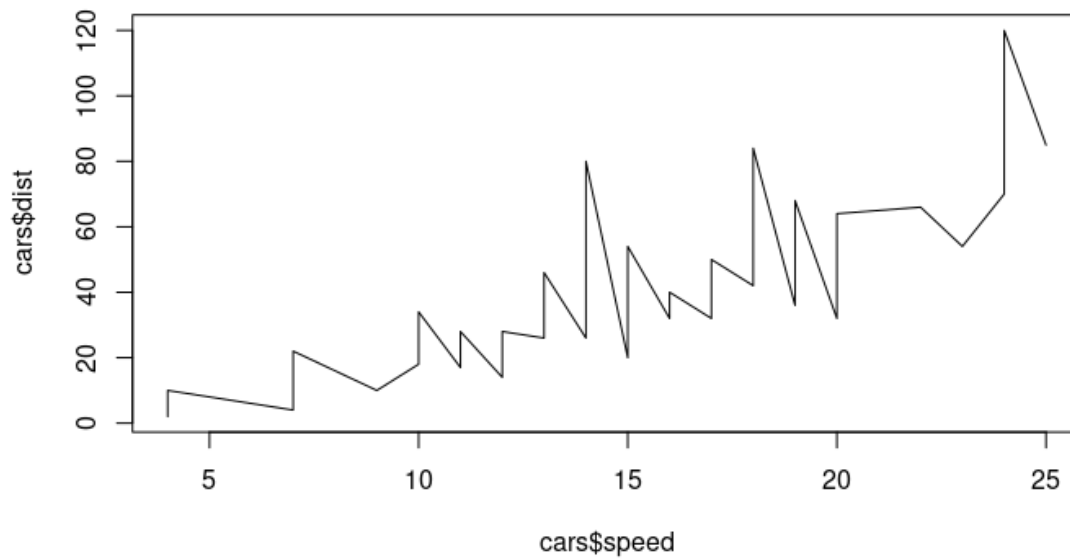
OUTPUT:





Histogram of cars\$speed





VIVA QUESTIONS

1. What is R?

R is a programming language used for statistical computing, data analysis, and graphical representation of data in various fields such as research and business analytics.

2. How do you assign a value in R?

In R, values can be assigned to variables using the assignment operator which is commonly written as an arrow or an equal sign, and the arrow operator is generally preferred in practice.

3. How to print a variable in R?

A variable can be printed in R by simply typing its name or by using the print function which explicitly displays the stored value.

4. What is a variable?

A variable is a named storage location used to hold data values that can be used and modified during program execution.

5. What does class function do in R?

The class function in R is used to determine the data type of a variable such as numeric, character, or logical.

6. What are the basic data types in R?

The main data types in R include numeric for numbers, character for text, logical for true or false values, and factor for categorical data.

7. What is a comment in R?

A comment in R is a line of text that is ignored during execution and is used to explain the code for better understanding.

8. What is a logical value in R?

A logical value represents truth values and can either be true or false in R.

9. Difference between arrow operator and equal operator?

Both operators are used for assignment in R but the arrow operator is more commonly used and recommended.

10. What is an R script?

An R script is a file that contains a sequence of R commands which can be executed together.

11. What is a vector?

A vector is a basic data structure in R that contains elements of the same data type arranged in a sequence.

12. How to create a vector?

A vector is created using the combine function which groups elements together into a single object.

13. What is vector addition?

Vector addition in R is performed element by element where corresponding elements from two vectors are added together.

14. What is recycling rule in R?

The recycling rule in R means that if vectors are of unequal length the shorter vector is repeated to match the length of the longer vector.

15. What is a matrix?

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A matrix is a two dimensional data structure consisting of rows and columns that store elements of the same type.

16. How to create a matrix?

A matrix is created using the matrix function by specifying the data and the number of rows or columns.

17. Difference between element wise and matrix multiplication.

Element wise multiplication uses the multiplication operator and works on corresponding elements while matrix multiplication follows linear algebra rules.

18. How to access elements in a matrix?

Elements of a matrix can be accessed by specifying the row and column positions inside square brackets.

19. What is an array?

An array is a multi dimensional data structure that can store data in more than two dimensions.

20. How to create an array?

An array is created using the array function by defining the data and dimensions.

21. What is subsetting

Subsetting refers to extracting specific elements from a data structure such as a vector or matrix.

22. How to subset a vector

Vector subsetting is done by specifying the index position inside square brackets.

23. How to subset a matrix

Matrix subsetting is done by specifying both row and column indices separated by a comma.

24. What is a list?

A list is a flexible data structure in R that can store elements of different data types in a single object.

25. How to create a list?

A list is created using the list function which can include numbers, strings, vectors, or even other lists.

26. How to access elements in a list?

Elements of a list can be accessed using either the dollar operator or double square brackets.

27. What is a factor?

A factor is a data type used to represent categorical variables with a fixed number of possible values.

28. What is an ordered factor?

An ordered factor is a type of factor where the levels have a meaningful order defined by the user.

29. What is a data frame?

A data frame is a table like structure in R where each column can contain different data types.

30. How to create a data frame?

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A data frame is created using the data frame function by combining vectors of equal length.

31. What is cbind function?

The cbind function is used to combine objects column wise into a single data structure.

32. What is rbind function?

The rbind function is used to combine objects row wise into a single data structure.

33. What does sort function do?

The sort function is used to arrange values in ascending or descending order.

34. How to extract a column from data frame?

A column from a data frame can be extracted using the dollar operator or by specifying the column name inside brackets.

35. How to check type of object in R?

The class function is used to check the type of an object in R.

36. What is an if statement?

An if statement is a control structure used to execute a block of code based on a condition.

37. What is the syntax of if statement?

The syntax of an if statement includes a condition followed by a block of code that executes if the condition is true.

38. What is a for loop?

A for loop is used to repeat a block of code for a fixed number of times.

39. What is a while loop?

A while loop is used to repeat a block of code as long as a specified condition remains true.

40. Difference between for and while loop.

A for loop runs for a fixed number of iterations while a while loop continues based on a condition.

41. What is a function?

A function is a reusable block of code designed to perform a specific task.

42. How to create a function in R?

A function is created using the function keyword followed by parameters and a block of code.

43. What is recursion?

Recursion is a technique where a function calls itself to solve smaller parts of a problem.

44. What is factorial?

The factorial of a number is calculated as the product of all positive integers up to that number.

45. What is Fibonacci series?

The Fibonacci series is a sequence where each number is the sum of the two preceding numbers.

46. What is quick sort?

Quick sort is a divide and conquer algorithm that sorts elements by selecting a pivot and partitioning the data.

47. What is binary search?

Binary search is an efficient searching algorithm that works only on sorted data and repeatedly divides the search space.

48. Difference between lapply and sapply.

lapply returns a list while sapply simplifies the result into a vector or matrix whenever possible.

49. What is a pie chart?

A pie chart is a circular graphical representation used to display proportions of a whole.

50. What is a histogram?

A histogram is a graphical representation used to show the distribution of numerical data by dividing it into intervals.